Sophie Yifei Wang

Undergraduate Researcher

AI Agents · Multimodal Learning · Human-centered AI

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Undergraduate in the Xu Teli Honors Program at Beijing Institute of Technology, majoring in Computer Science and Technology. My research interests include **AI agents, long-term memory, multimodal large language models, model interpretability, and human-computer interaction**. I am particularly interested in translating emerging model capabilities into efficient, transparent, and human-centered intelligent systems.

Education

Beijing Institute of Technology
Sep 2024 – Present

B.Eng. Candidate in Computer Science and Technology · Xu Teli Honors Program

- Ranked in the top 15% of the major and qualified for recommended postgraduate admission; class representative of the Xu Teli Honors Program.
- CET-4 score: 572; Second Prize in the National English Competition for College Students.

Research Experience

Threader · Long-term Memory for Conversational Agents
Dec 2025 – May 2026

Research collaboration with Tsinghua University · Co-author · NeurIPS 2026 under review

- Co-developed Threader, a lossless long-term memory framework addressing information loss, latency, and cost in compression-based memory systems.
- Introduced a topic segmentation, multi-view encoding, and multi-signal retrieval pipeline that preserves complete conversations while improving construction and retrieval efficiency and reducing token cost.
- Outperformed representative systems including MemGPT and Mem0 on long-conversation memory benchmarks, improving long-range consistency and response efficiency.

SAIL · Brain-Model Semantic Alignment
Sep 2025 – May 2026

Beijing Institute of Technology · Co-author · NeurIPS 2026 under review

- Co-developed a sparse autoencoder (SAE) framework for aligning brain activity with interpretable representations inside multimodal foundation models.
- Disentangled high-dimensional visual features into monosemantic concepts and matched them to fMRI signals from the human visual cortex, revealing hierarchical correspondences with the ventral visual pathway.
- Demonstrated that SAE features outperform raw model features in brain-alignment tasks.

Industry Experience

ByteDance · Volcano Ark
Apr 2026 – Jun 2026

Product Solutions Intern · AI Products & MaaS Solutions

- Contributed to enterprise AI product and MaaS solution design by mapping foundation-model capabilities to business scenarios and product delivery paths.
- Supported solution research and product communication for B2B AI use cases, translating technical capabilities into actionable business proposals.

Selected Projects

Collector+ (Collector-AI)
Jan 2026

FastAPI · React · LLM · Prompt Engineering

- Built a low-latency information extraction and gamified reading assistant that transforms long-form articles and videos into binary predictive questions and supports source-grounded Q&A over saved content.
- Designed a linear monolithic asynchronous pipeline with approximately 15–30 seconds end-to-end generation latency and universal parsing for WeChat articles and Bilibili videos.
- GitHub · Live Demo

FitSnap

Dec 2025

Next.js 16 · React 19 · DeepSeek-V3 · Zustand

- Transformed unstructured instructional videos into structured, timestamped steps with action categories and key parameters, with bidirectional synchronization between video playback and step cards.
- Built a content-agnostic generative UI that dynamically renders domain-specific components for fitness, cooking, and other instructional content.
- Live Demo

BiGEM Data Retrieval and Visualization Platform

Sep 2024 – Oct 2025

React · D3.js · Three.js · Python

- Cleaned and connected iGEM data from 2004–2024 and visualized global collaboration networks with force-directed graphs, trees, and Sankey diagrams.
- Integrated LLM-based semantic analysis to improve retrieval of biological parts and historical projects; received a 2025 iGEM Gold Medal and Best Software in the undergraduate category.
- Source · Project Wiki

Multi-Agent Shopping Assistant

Nov 2025 – Present

Python · Vue 3 · MCP · RAG

- Built an e-commerce assistant around the Model Context Protocol, implementing the LLM → Router → MCP Server → Backend API tool-calling pipeline.
- Developed a Vue 3 conversational interface that renders tool results as visual product cards.
- GitHub

Honors & Awards

iGEM Competition · Gold Medal and Best Software, Undergraduate Category	2025
China Undergraduate Mathematical Contest in Modeling · Beijing First Prize	2025
Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling · Honorable Mention	2025
National English Competition for College Students · Second Prize	2025
Beijing Institute of Technology · First- and Second-Class Scholarships	2025

Technical Skills

AI Engineering	PyTorch, Prompt Engineering, Structured Output, Function Calling, RAG, Long-context Management
Frontend & Visualization	Next.js, React, Vue 3, TypeScript, D3.js, Three.js, Framer Motion
Backend & Engineering	Python, FastAPI, Git, Docker, Linux, Data Pipelines
Languages	Chinese (Native), English (CET-4 572; Professional working proficiency)

Last updated: July 2026